

WiFi Indoor Localization with Deep Learning

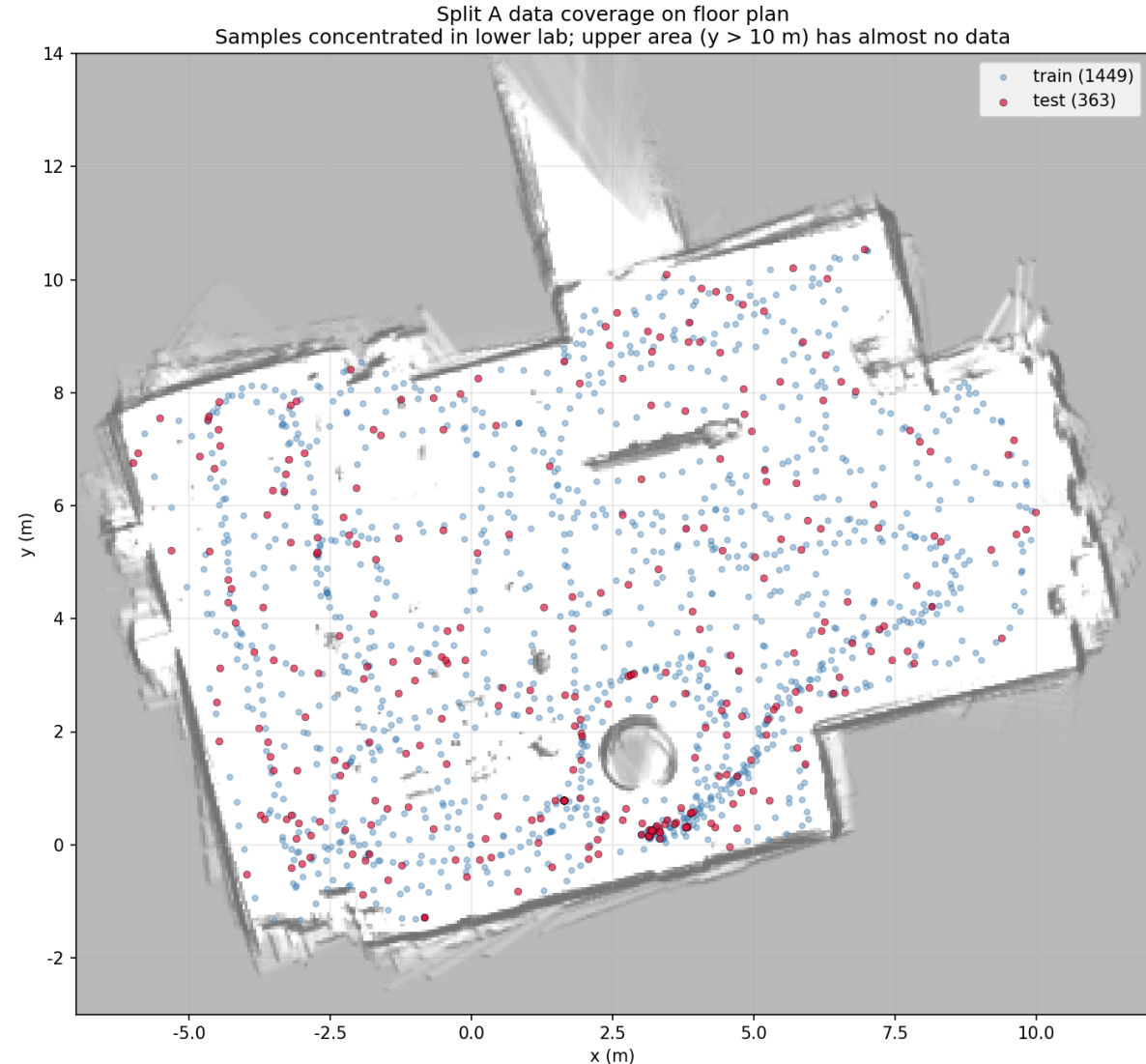
From a 1.57 m KNN baseline to a 0.79 m cascade model

The task: turn one WiFi scan into a point on the map

- Input: one scan = a set of (BSSID, RSSI) pairs
- Output: robot position (x, y) in metres on the floor plan
- No GPS indoors; WiFi fingerprinting is the classic fallback

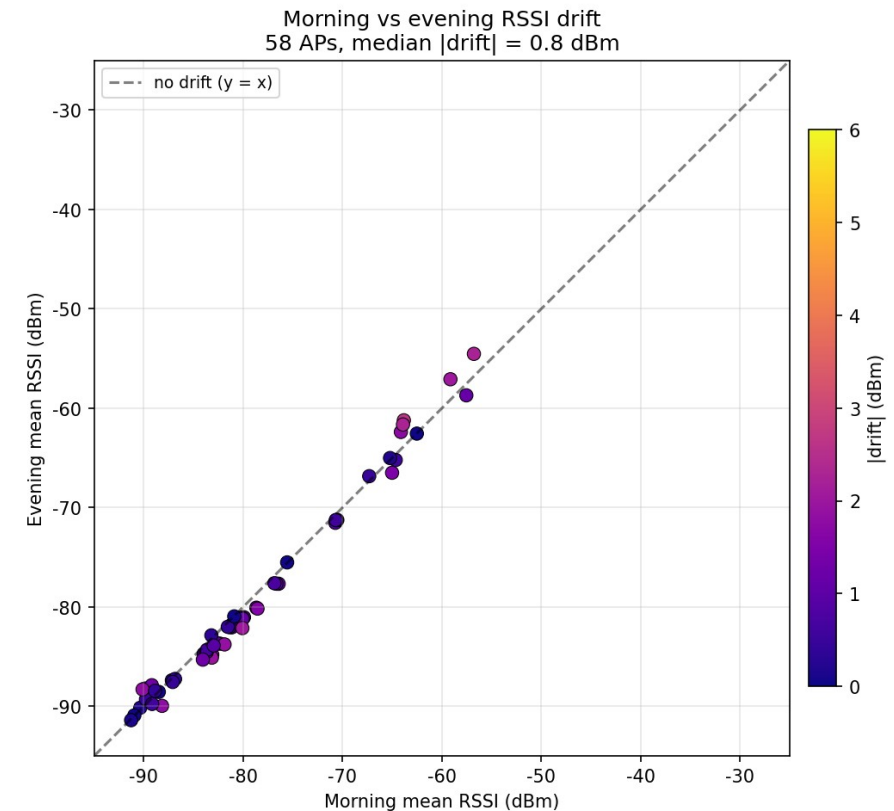
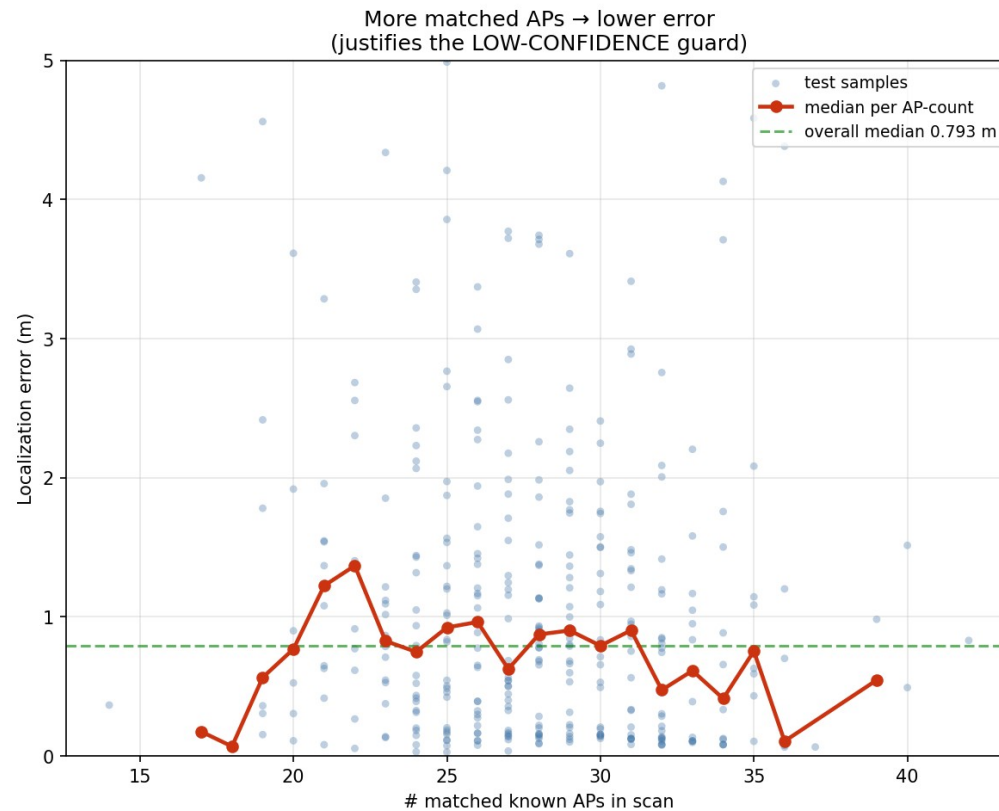
1,812 fingerprints, 189.5 m² lab, two sessions

- Morning 912 + evening 900 (RSSI, pose) records
- 115 BSSIDs seen; 80 kept (≥ 10 hits), rest are noise
- Coverage follows the driven path; upper region barely visited



Coverage is the prime suspect, not noise

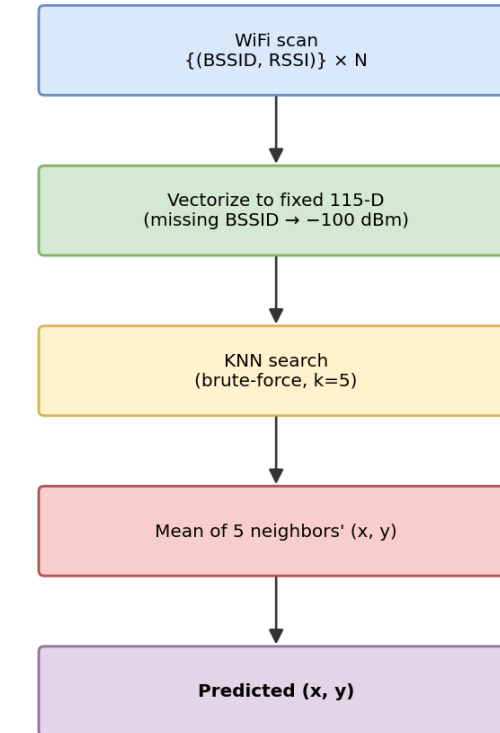
- Within-cell RSSI std ~ 3.5 dBm
- Morning vs evening median drift ~ 0.8 dBm
- Error rises when few APs match, then flattens



Baseline: classic KNN fingerprinting lands at 1.57 m

- Vectorize scan to 80-D RSSI; missing AP = -100 dBm
- k=5 nearest neighbours in RSSI space, distance-weighted pose
- No training; the baseline every neural net must beat

KNN k=5 (baseline, median 1.568 m)

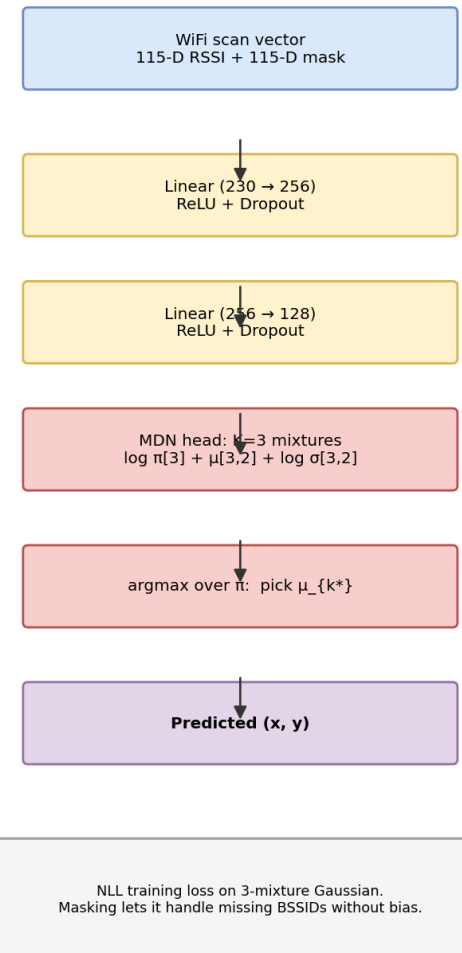


No training. Distance metric = Euclidean in 115-D RSSI space. Test ⇒ median 1.568 m.

Plain nets barely beat KNN

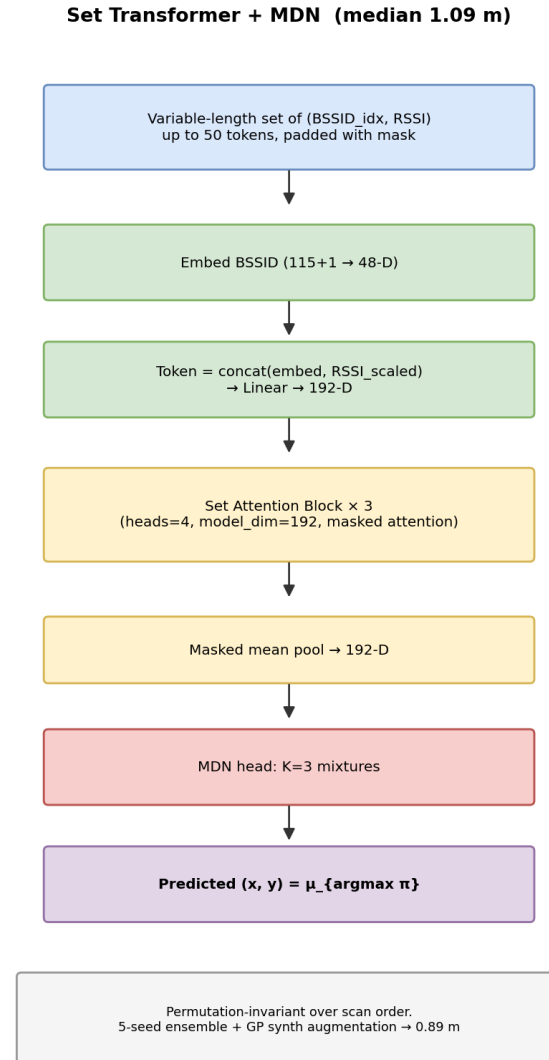
- MLP regression: 1.30 m, only marginally below KNN
- MaskedMDN: 1.37 m, no gain over the MLP
- Masking a dense, -100-padded vector does not help

MaskedMDN (NN baseline, median 1.20 m)



Treat the scan as a set, not a vector: 1.09 m

- Each AP = one token: BSSID embedding + scaled RSSI
- Set Transformer, masked + permutation-invariant attention
- Only real APs enter — no -100 padding

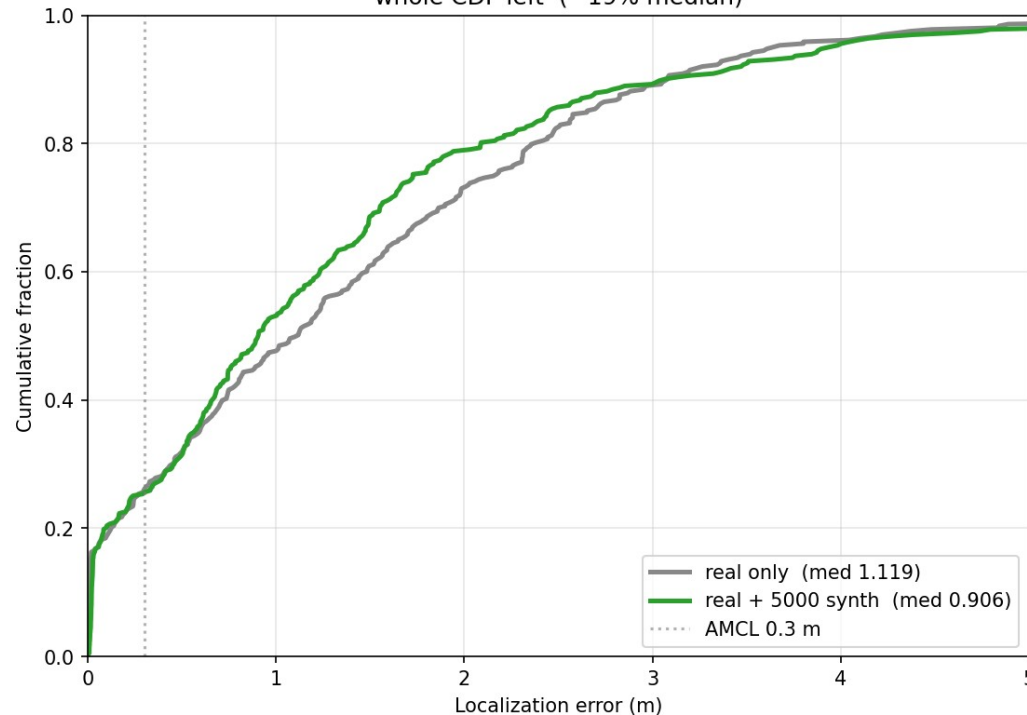


Synthesize coverage where data is sparse: -19% in-session (1.12 -> 0.91 m)

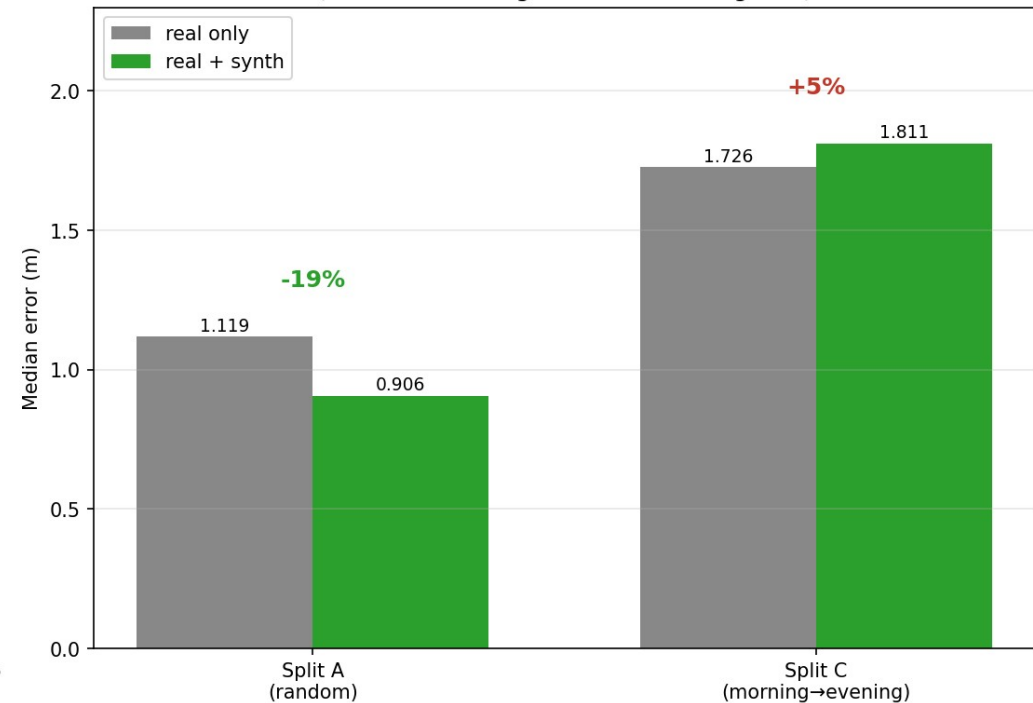
- Ablation vs a real-only baseline (1.12 m), not vs the 1.09 m set model
- Per-BSSID Gaussian Process fills free space
- Split A: -19% (1.12 -> 0.91 m); Split C: +5%, worse
- Kept synth: gain is in-session; cross-session drift is revisited at the end

GP-kriging synthetic data: fills the spatial coverage gap (1449 real positions → +5000 free-space samples)

Split A (random): synthetic data shifts the whole CDF left (-19% median)



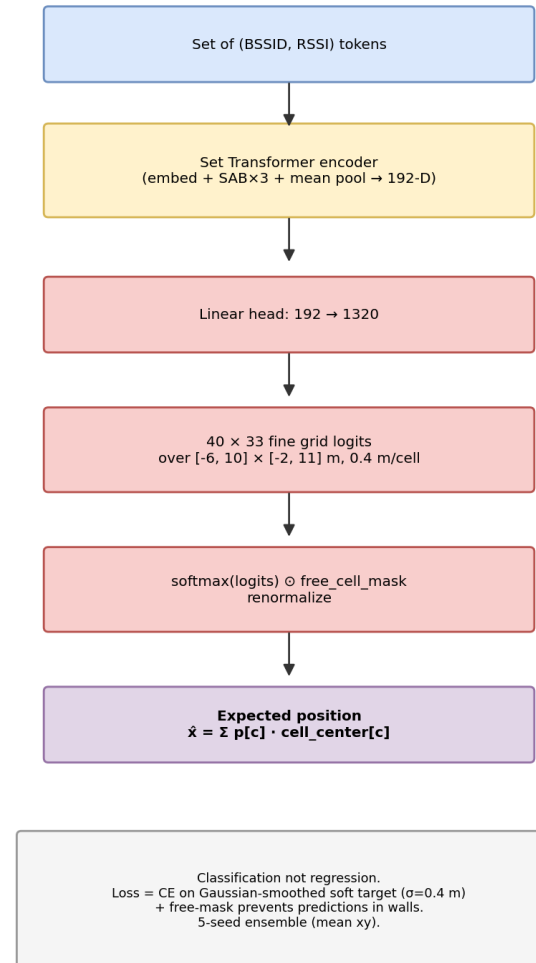
Helps on random split, slightly hurts on transfer (GP fit on morning → carries morning bias)



Classifying on a grid beats averaging: 0.88 m

- Split floor into ~ 0.4 m cells
- Predict probability per cell, not a coordinate
- Mask zeroes walled cells

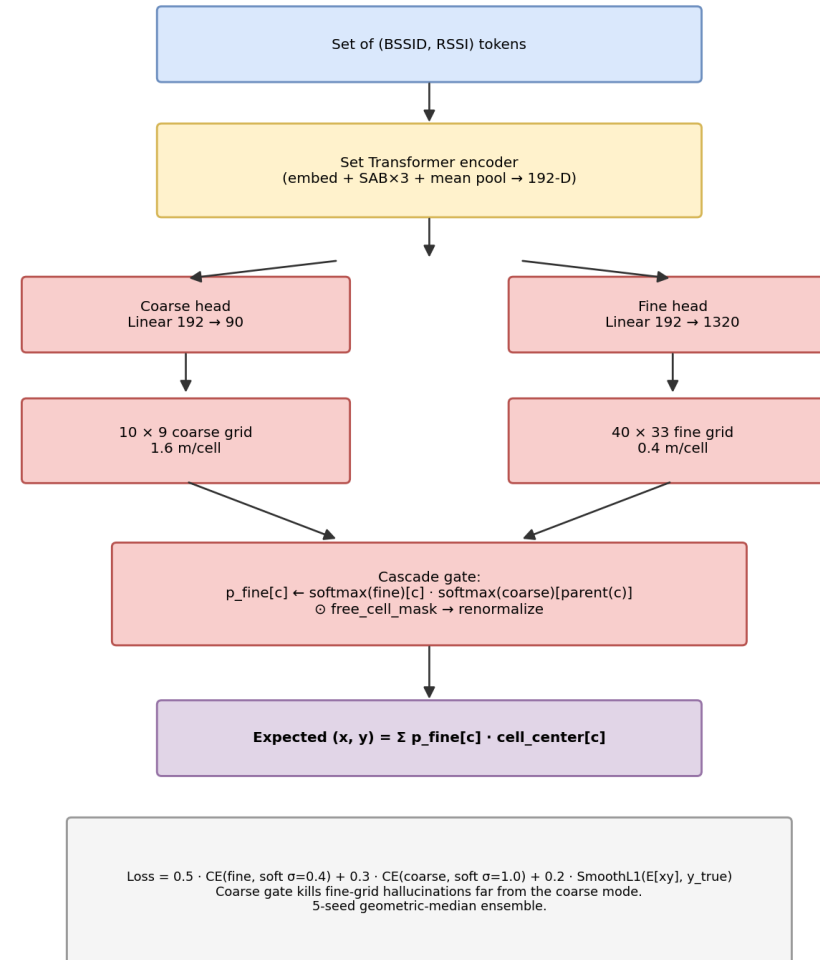
Set Transformer + Heatmap (median 0.836 m, $\times 5$ -ens)



Coarse-gate the fine grid: 0.79 m

- Shared encoder, two heads: coarse + fine grid
- Gate = fine prob x parent coarse-cell prob
- 0.79 m median, 5-seed ensemble

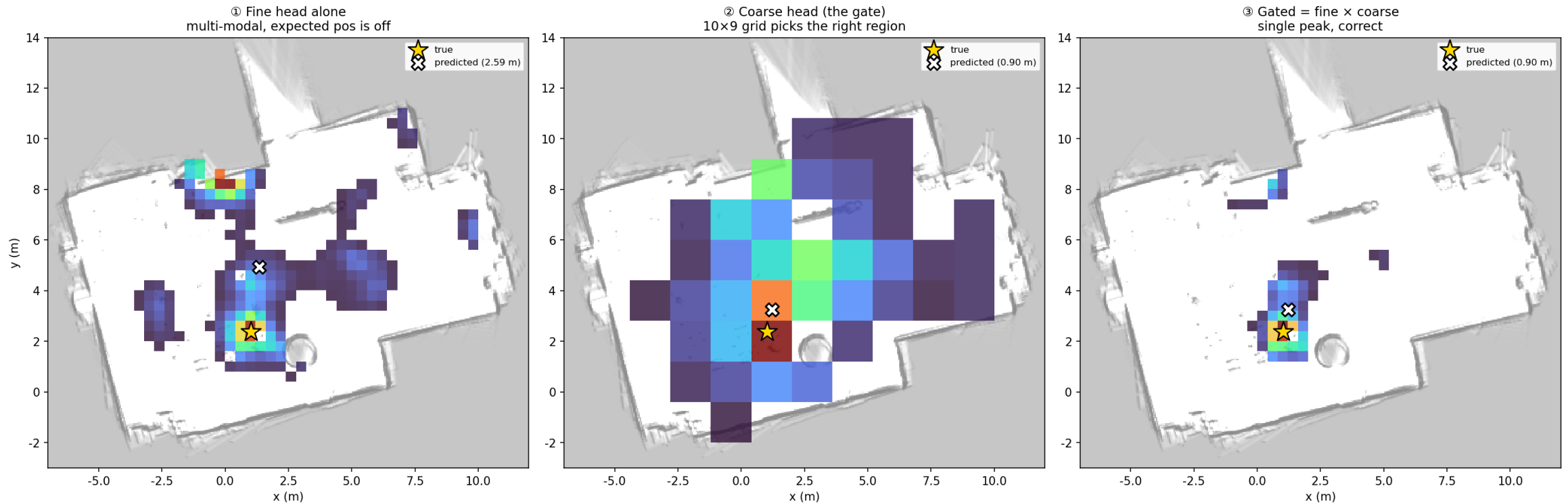
Coarse-to-Fine Cascade [WINNER, median 0.793 m, x5-ens]



Why it works: the coarse gate suppresses the fine grid's false peak

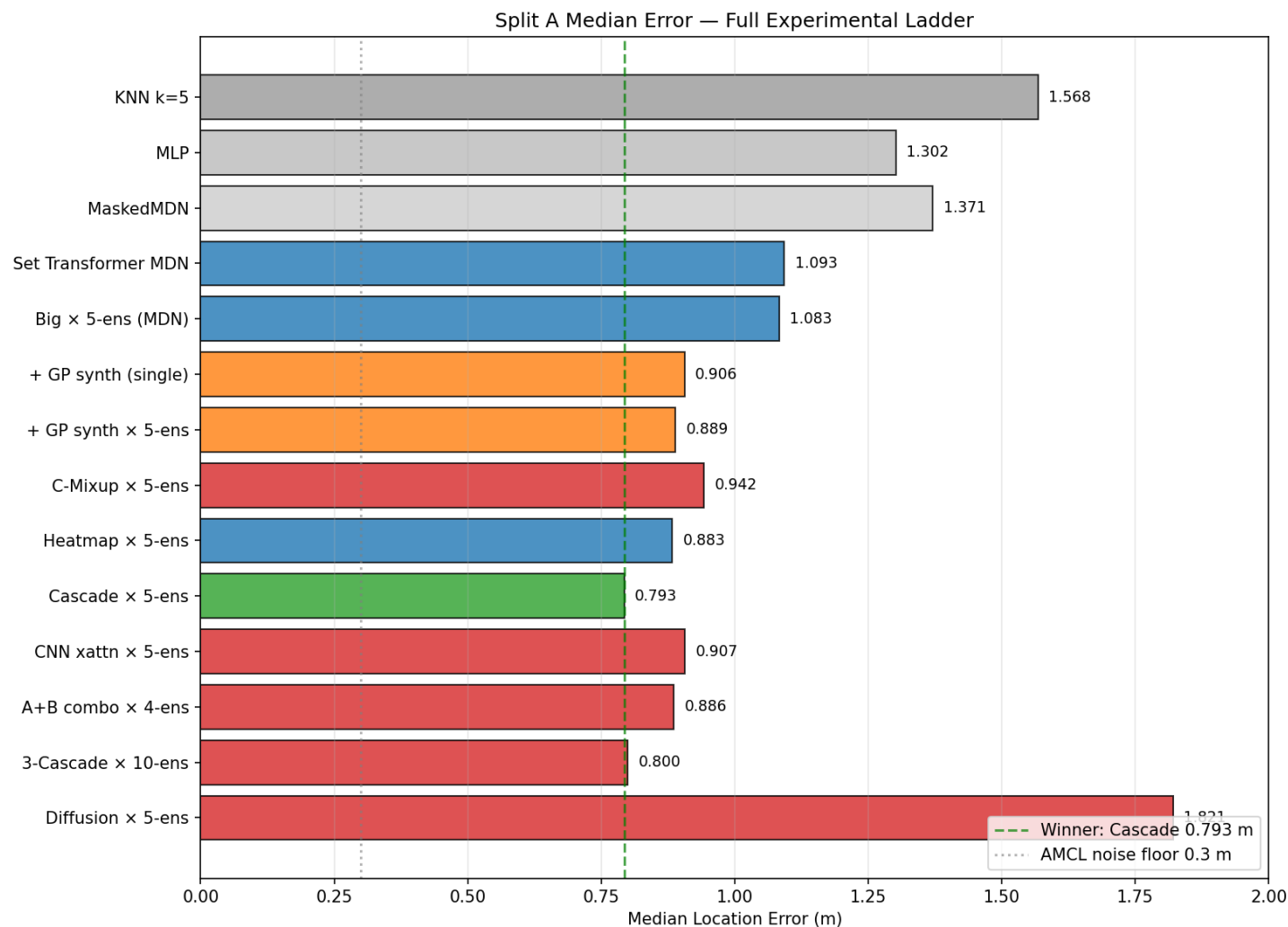
- Fine head alone: two peaks
- Expectation lands between them, off target
- Coarse gate keeps one peak, on target

Why the cascade wins: the coarse gate removes fine-grid ambiguity (this sample: 2.59 m $\rightarrow$ 0.90 m)



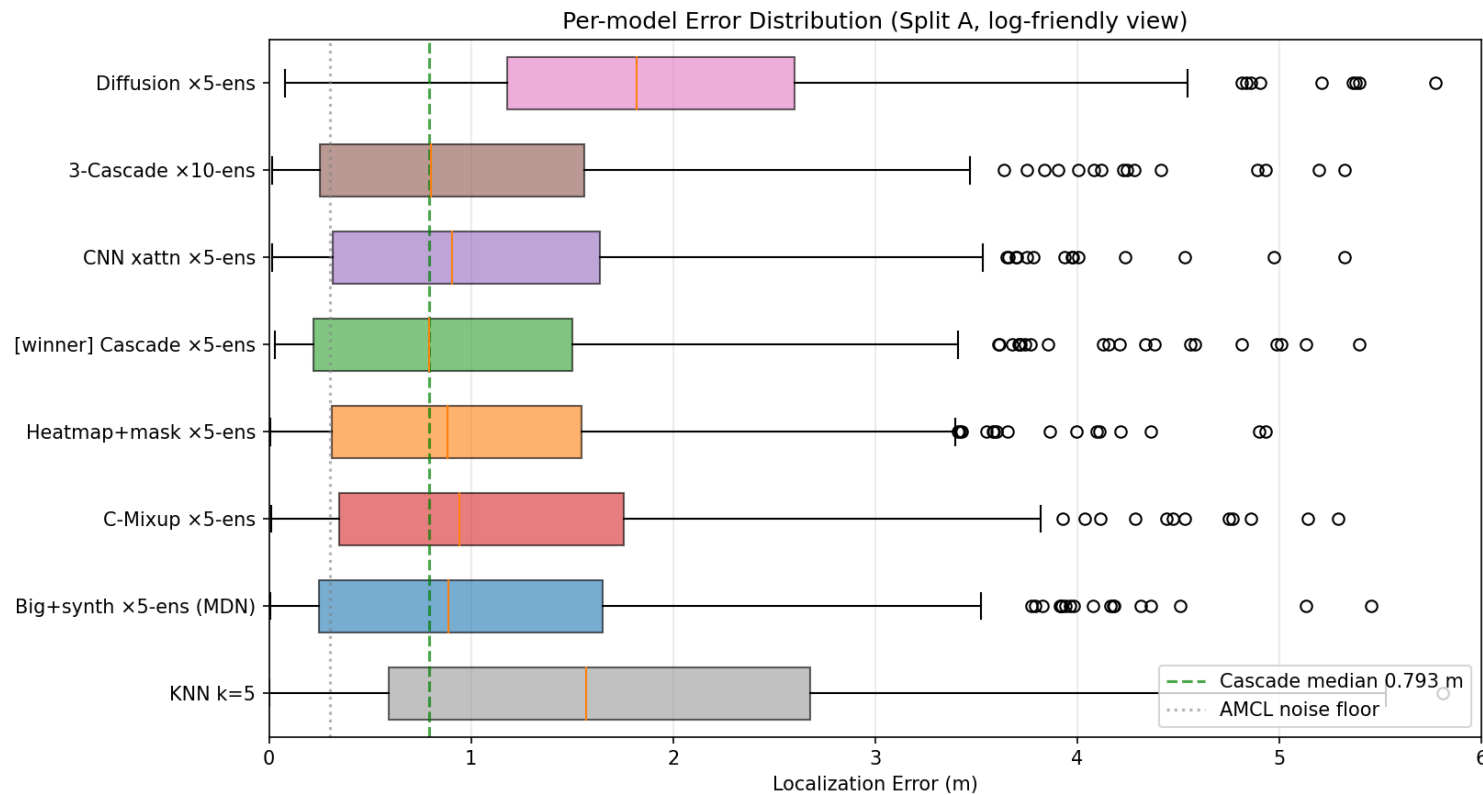
The full climb: 1.57 m -> 0.79 m across the experiments

- Each bar is one experiment on the same Split A test set
- Green = winner, red = abandoned
- From set-transformer onward, milestones are 5-seed ensembles; KNN/MLP/MaskedMDN are single models
- Largest gains: the set representation and synthetic coverage



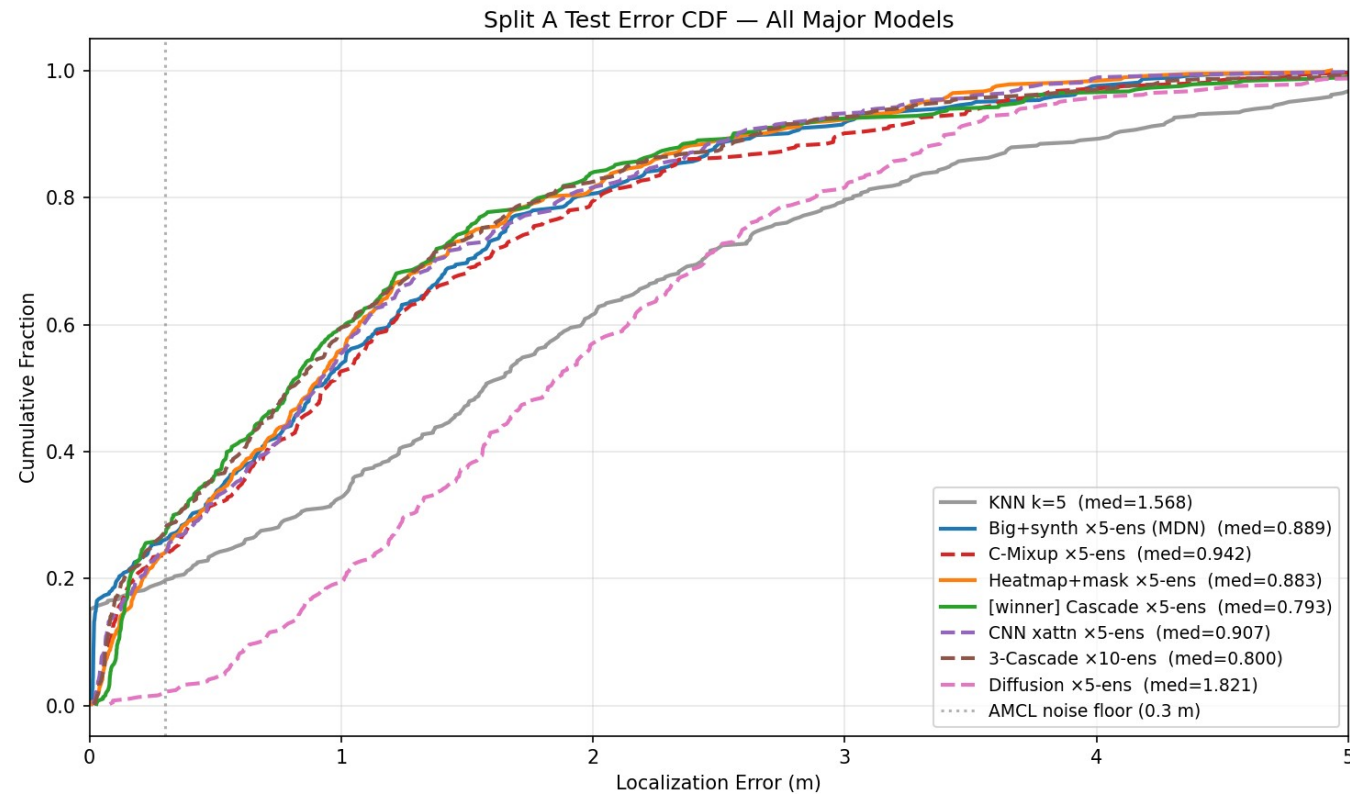
Larger and more complex variants did not beat the cascade

- Diffusion: 1.82 m, adds sampling noise
- 3-level cascade: 0.80 m, no gain at 2x ensemble cost
- CNN cross-attention / C-Mixup: 0.91-0.94 m



Error distribution: cascade is best at every percentile

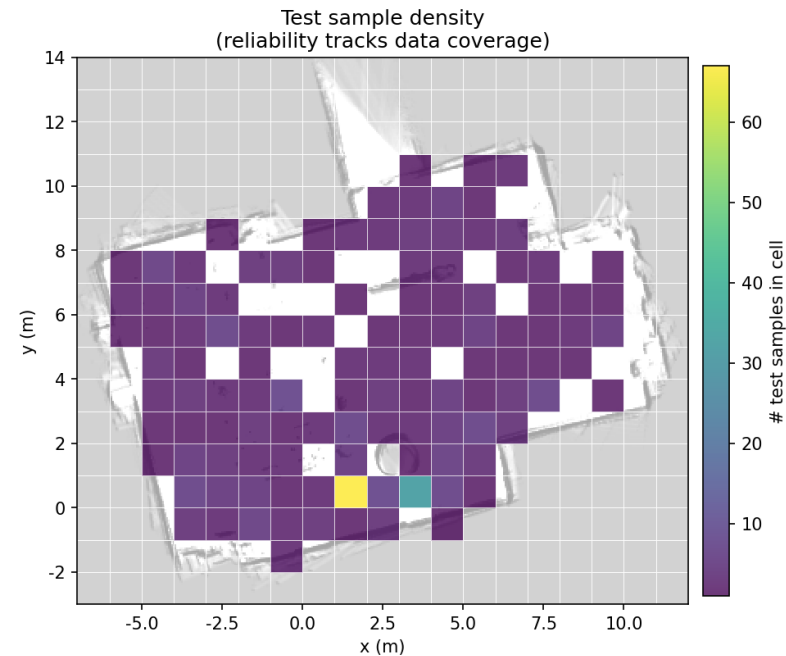
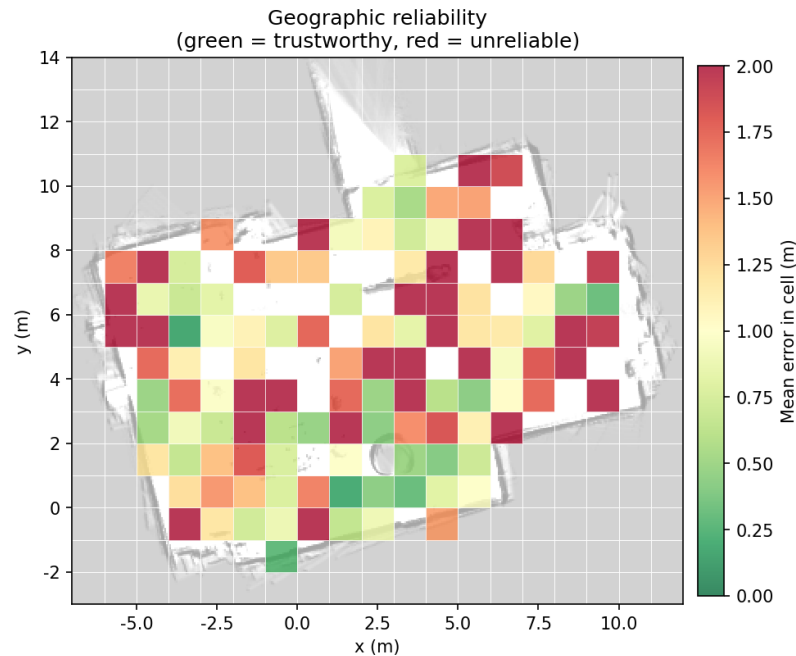
- Median 0.79 m, mean 1.12 m, p90 2.56 m
- 27% of predictions within 0.3 m
- >2 m tail likely from sparse-region / symmetric-layout ambiguity



Where to trust it: reliability tracks data density

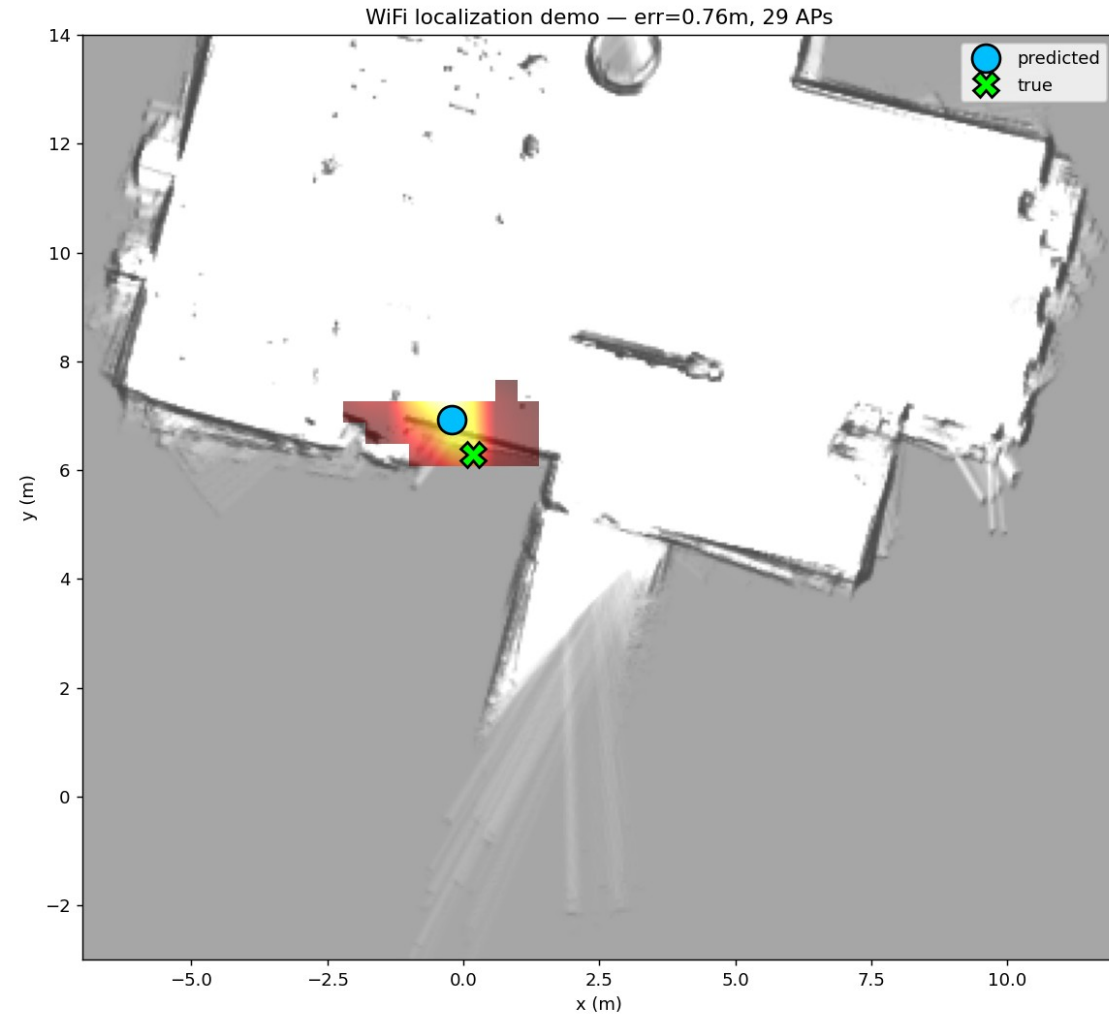
- Per-cell mean error mirrors the sample-density map
- Dense, well-travelled regions: well under 0.5 m
- Sparse upper region: error climbs above 2 m
- Fix is more data, not a bigger model

Where can you trust the model? Error is low where data is dense, high in the sparse upper region.



Live in the lab: predictions consistent with the offline median

- ESP32 streams scans over USB; ROS 2 node + RViz visualisation
- 80-95% of scanned APs match the trained vocabulary
- Standing still: prediction spread under 0.1 m
- Single ground-truth check: 0.4-0.8 m, consistent with the 0.79 m median



The one unsolved gap: time-of-day drift

- Cross-session morning -> evening still degrades (synth +5%)
- No heading in the fingerprint
- ~3.8 s scan update period

From 1.57 m to 0.79 m, and proven on real hardware

- Halved the KNN error: 1.57 m $\rightarrow$ 0.79 m median
- Classification + a coarse gate beat direct regression
- Reliability is bounded by data coverage, not the network
- Next gain: handling cross-session drift, not adding depth